

Performance Analysis of Load Frequency Control for a Two-Area Interconnected Power System using Twisting Control

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OUTLINE

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INTRODUCTION

The load frequency control (LFC) is recognized as the automatic generation control (AGC) deals with the **frequency** and **active power** .

Need to bring the value of area control error (ACE) to zero.

- Integral controllers
- Higher order sliding mode (HOSM) control

HOSM exhibit better prospects for

- Fast and finite time convergence.
- High precision control.

Normally used HOSM controllers are **twisting** and **super-twisting** controllers.

LITERATURE SURVEY

1. Sliding Mode Based Load Frequency Control for a Two-area Interconnected Power System- Jianping guo and Lili dong

- The paper compares sliding mode based LFC with the conventional PI based LFC.
- The simulation results show the superiority of the sliding mode based LFC to PI based LFC in terms of settling time and overshoot percentage.

2. Discrete-time twisting controller without numerical chattering: analysis and experimental results- O. Huber, V. Acary, B. Brogliato and F. Plestan

- This paper presents the results of an experimental study of two discrete-time twisting controllers: the implicit and the explicit one.
- It shows the theoretical superiority of the implicitly discretized version.

3. Load frequency control of interconnected five-area power system with pid controller- Avinash Sahu and Dr. L.B.Prasad

- In this paper load frequency control is done by PID controller.
- This paper for single area, two area and five area the helpfulness of PID controller is confirmed.

PROBLEM AND OBJECTIVE

PROBLEM FORMULATION

- Optimal control of frequency, tie-line power flow and Area Control Error (ACE) of two area interconnected power system need to be solved using appropriate controller in the presence of disturbance.

DETAILED OBJECTIVE

- Design of **Twisting** sliding mode controller (SMC) for optimal load frequency control (LFC) of two area interconnected power system.
- **Comparison** of higher order sliding mode (HOSM) controller with existing methods.

BLOCK DIAGRAM

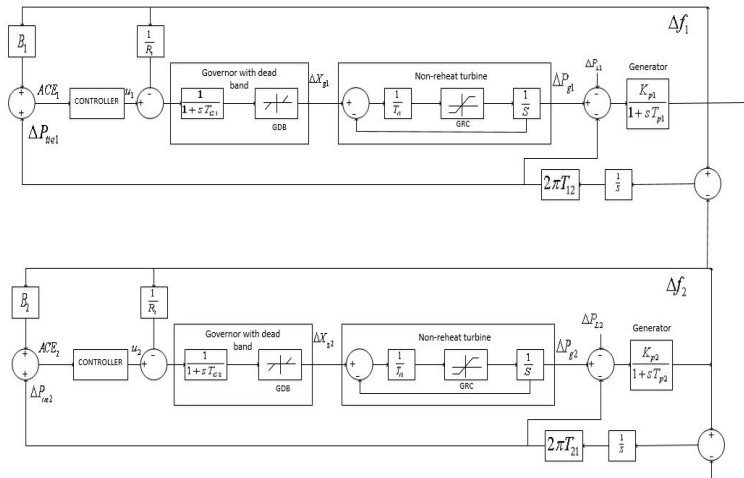


Figure: 1

Fundamental load frequency control loops

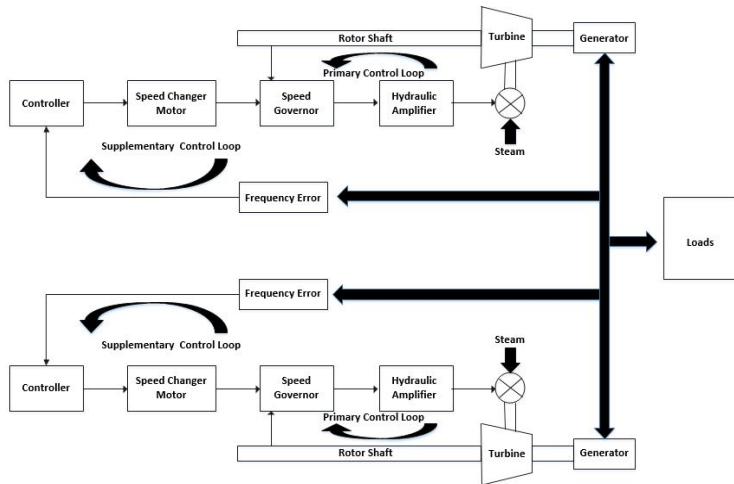


Figure: 1

PRIMARY CONTROL

- There are mainly three types of errors such as frequency error, tie-line power error and area control error (ACE).
- When the error is large governor action is introduced and the generator output adjusted to nullify the effect of load disturbance. This type of governor action is called the **primary control**.
- The governor action will bring the frequency near to standard value. But it does not ensure the desired value based on other interconnected systems.

SECONDARY CONTROL

Objectives

- Hold **system frequency** at or very close to a specified nominal value.
 - Maintain correct value of **interchange power** between control areas
 - Maintain each units generation at the **most economic** value.
-
- The required change in generation is called **Area Control Error(ACE)**
 - ACE represents the shift in the areas generation required to restore frequency and net interchange to their desired values

The equation for ACE for each areas:

$$ACE_1 = -\Delta P_{net-int1} - B_1 \Delta \omega$$

$$ACE_2 = -\Delta P_{net-int2} - B_2 \Delta \omega$$

B_1, B_2 are called **Frequency Bias Factors**

SECONDARY CONTROL-continue

Supplementary control action

- Once a load change occurred, a supplementary control must act to restore the frequency to nominal value.
- This can be done by adding a reset control to the governor.

Normally used reset controller is **Integral Controller**

- Reset control action will force the frequency error to zero by adjustment of the reference set point.

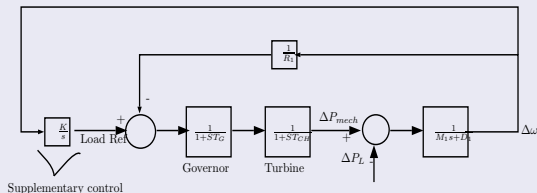


Figure: 10

SLIDING MODE CONTROL

- There are two main **advantages** of SMC
 - 1) **Dynamic behaviour** of the system can be controlled by the particular choice of the sliding function.
 - 2) The **closed loop response** becomes totally insensitive to some particular uncertainties which are bounded.

From practical point of view SMC allows for controlling nonlinear processes subject to **external disturbances** and **heavy model uncertainties**.

- SMC consist of two phases

Phase 1 :Sliding surface design

Phase 2 :Control input design

SLIDING MODE CONTROL-Continue

PHASE 1 :SLIDING SURFACE DESIGN

- The sliding surface depends on the tracking error together with a certain number of its derivatives
ie, $\sigma = \sigma(e, \dot{e}, \ddot{e}, \dots, e^{(k)})$
- The function σ should be selected in such a way that it is vanishing.
- $\sigma = 0$, gives rise to a **stable** differential equation.
- Any solution $e_y(t)$ of which will tend to zero

Typical choice for the sliding manifold is

$$\sigma = \dot{e} + c_0 e$$

$$\sigma = \ddot{e} + c_1 \dot{e} + c_0 e$$

$$\sigma = e^{(k)} + \sum_{i=0}^{k-1} c_i e^{(i)}$$

SLIDING MODE CONTROL-Continue

PHASE 2 :CONTROL INPUT DESIGN

Finding a control action that directs the system trajectories onto the sliding manifold, or able to steer the σ variable to zero in finite time.

There are two parts in control input

- Bring the system trajectory to the sliding surface $\Rightarrow$ **Equivalent control**
- Maintain the system trajectory on the surface $\Rightarrow$ **Switching control**

Common feature of all sliding mode based techniques is that no precise information about the original system dynamics is requested, the controlled system being treated as a completely uncertain.

SLIDING MODE CONTROL-Continue

RESTRICTIONS

- Conventional SMC provide **robust and high-accuracy solutions** for a wide range of control problems under uncertainty conditions. However, **restrictions remain**.

Restriction-1

The constraint to be held at zero in conventional sliding modes of degree one, which means that the control needs to explicitly appear in the first time derivative of the constraint. Thus, one has to **search for an appropriate constraint**.

Restriction-2

High-frequency control switching may easily cause unacceptable practical complications (**chattering effect**).

- The chattering effect can degrade the system performance or may even lead to instability and may even damage the plant

- Using a **Second-order sliding mode technique**, the given task can be accomplished exactly, and in finite time, by means of **continuous control**. As a consequence it can significantly **reduce the chattering effect**.
- The twisting controller is the first second order sliding controller which was proposed.

formula

$$u = -r_1 \text{sign}(\sigma) - r_2 \text{sign}(\dot{\sigma}) \text{ where } r_2 \succ r_1$$

- The controller in the above equation guarantees the appearance of a second order sliding mode attracting the trajectories of the sliding variable dynamics in finite time to zero.

Block Diagram Realization

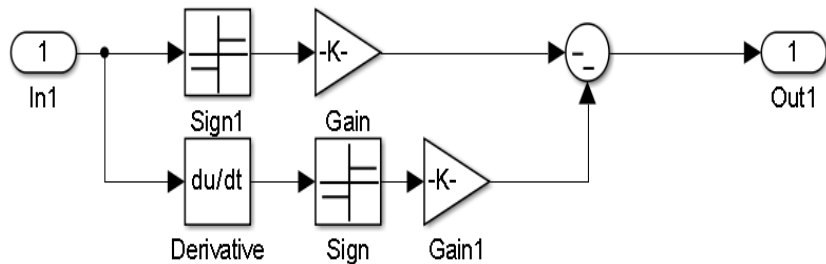


Figure: 14

Control input

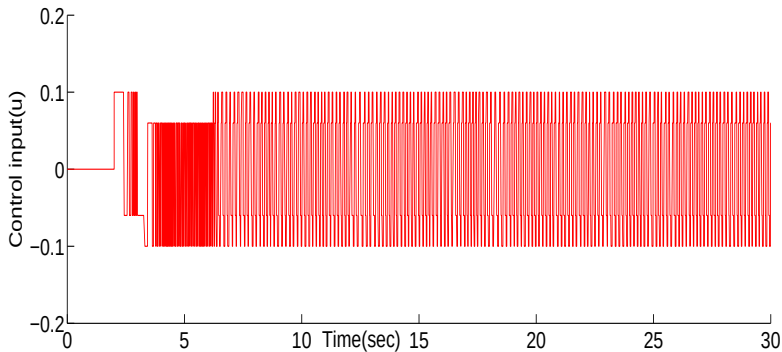


Figure: 15

SIMULATION PARAMETERS

The nominal system parameter values are listed as follows.

$$T_{g1} = 0.08s, T_{g2} = 0.09s$$

$$T_{t1} = 0.3s, T_{t2} = 0.4s$$

$$T_{p1} = 20s, T_{p2} = 25s$$

$$K_{p1} = 120\text{Hz/pu.MW}, K_{p2} = 130\text{Hz/pu.MW}$$

$$R_1 = 2.4\text{Hz/pu.MW}, R_2 = 2.6\text{Hz/pu.MW}$$

$$B_1 = 0.425\text{puMW/Hz}, B_2 = 0.44\text{pu.MW/Hz}$$

$$2\pi T_{12} = 0.545\text{pu.MW/rad}$$

$$2\pi T_{21} = 0.6\text{pu.MW/rad}$$

LOAD DISTURBANCE CONSIDERED FOR SIMULATION

Step pulse of 0.02pu magnitude with unit period applied at 2sec

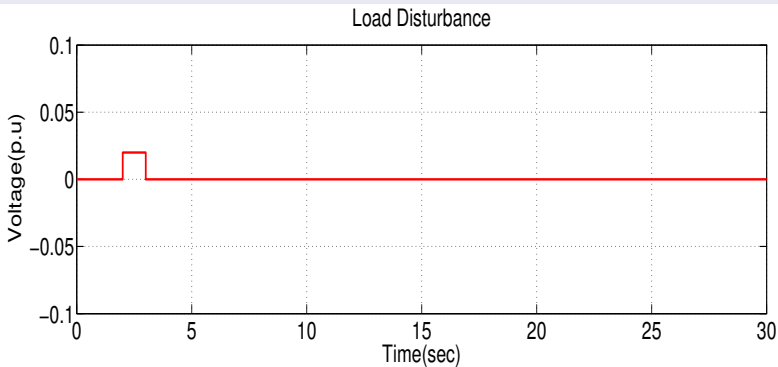
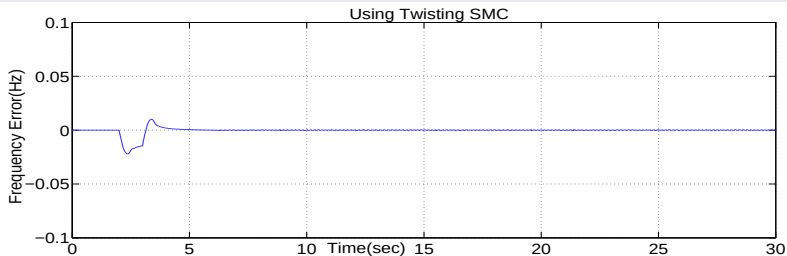
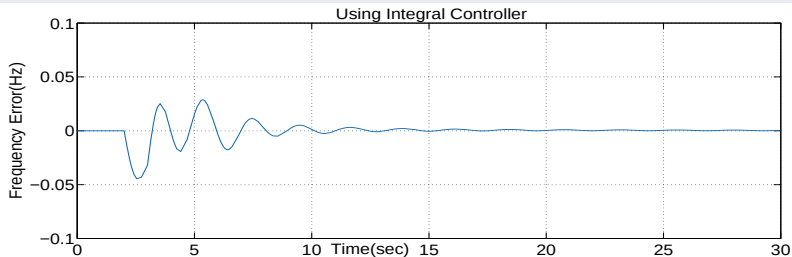


Figure: 19

Frequency v/s Time

Integral v/s Twisting

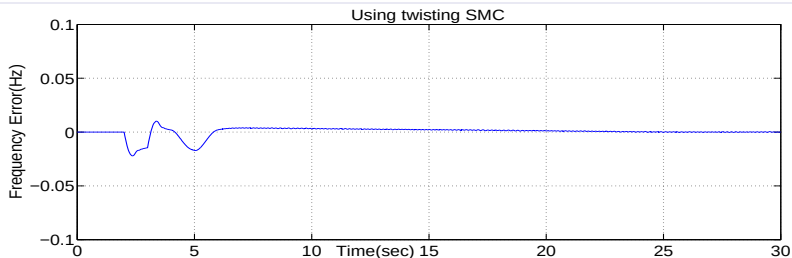
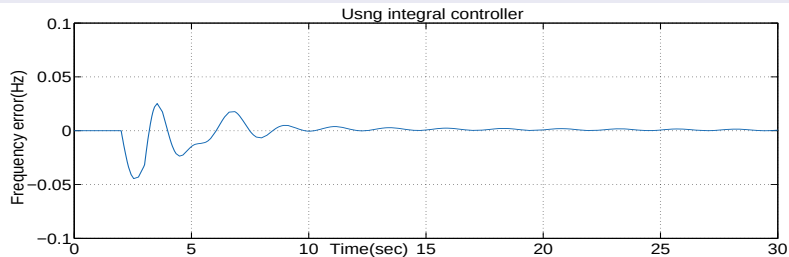
Disturbance applied to one area: Area-1 at 2sec



Frequency v/s Time-Continue

Integral v/s Twisting

Disturbance applied to both areas: Area-1 at 2sec and Area-2 at 4sec



PERFORMANCE ANALYSIS

From Frequency v/s Time graph

Table: Disturbance applied to **one area**

Case	Controller	Settling time	Overshoot	Undershoot
1	Integral	20	0.03	-0.04
2	Twisting SMC	4	0.01	-0.02

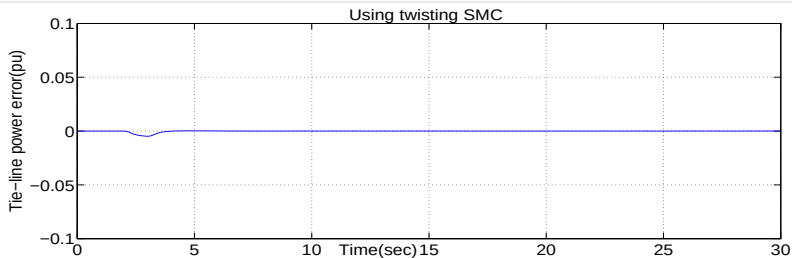
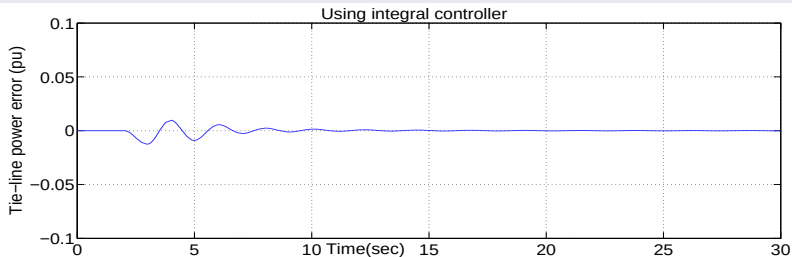
Table: Disturbance applied to **both areas**

Case	Controller	Settling time	Overshoot	Undershoot
1	Integral	20	0.025	-0.04
2	Twisting SMC	20	0.01	-0.02

Tie-line power v/s Time

Integral v/s Twisting

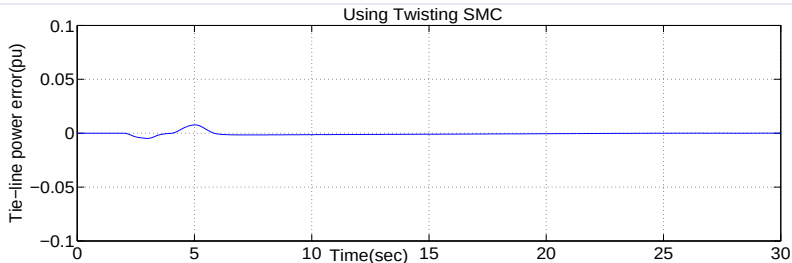
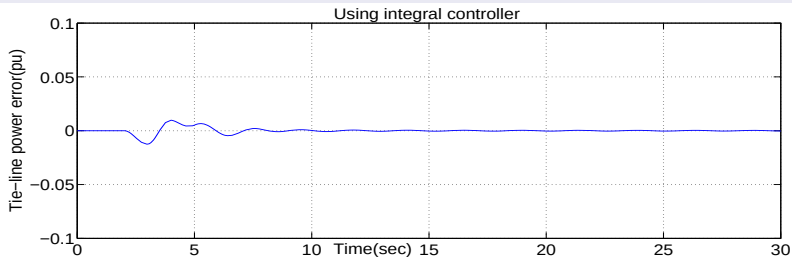
Disturbance applied to one area: Area-1 at 2sec



Tie-line power v/s Time-Continue

Integral v/s Twisting

Disturbance applied to both areas: Area-1 at 2sec and Area-2 at 4sec



PERFORMANCE ANALYSIS

From Tie-line power v/s Time graph

Table: Disturbance applied to **one area**

Case	Controller	Settling time	Overshoot	Undershoot
1	Integral	15	0.01	-0.015
2	Twisting SMC	3.5	0	-0.01

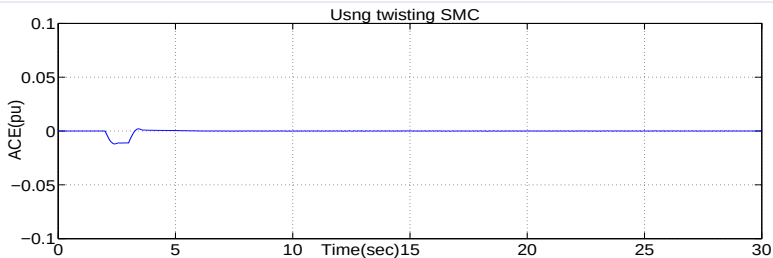
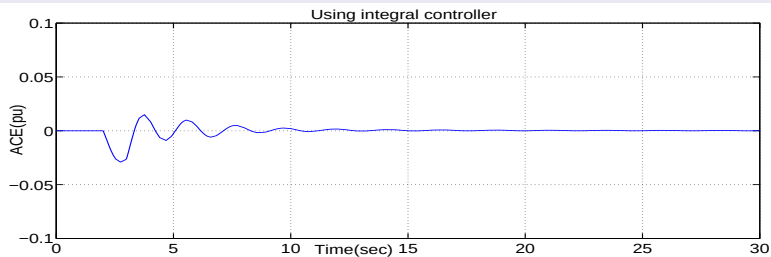
Table: Disturbance applied to **both areas**

Case	Controller	Settling time	Overshoot	Undershoot
1	Integral	9	0.01	-0.02
2	Twisting SMC	6	0.01	-0.005

ACE v/s Time

Integral v/s Twisting

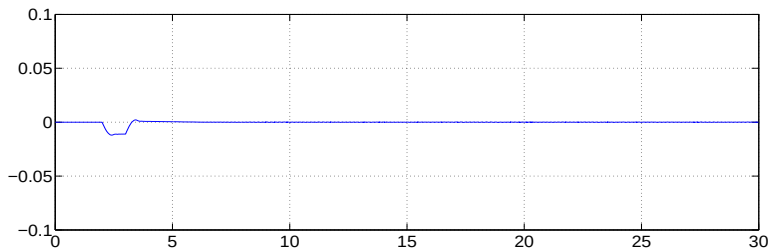
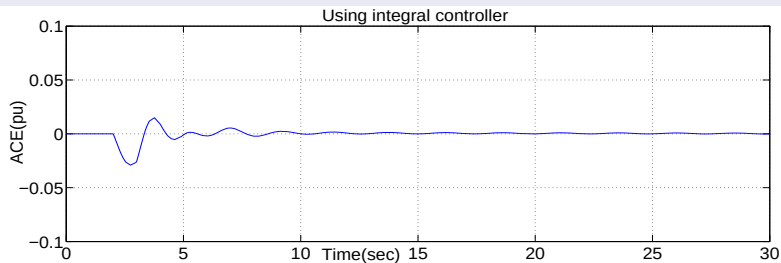
Disturbance applied to one area: Area-1 at 2sec



ACE v/s Time-Continue

Integral v/s Twisting

Disturbance applied to both areas: Area-1 at 2sec and Area-2 at 4sec



PERFORMANCE ANALYSIS

From ACE v/s Time graph

Table: Disturbance applied to **one area**

Case	Controller	Settling time	Overshoot	Undershoot
1	Integral	20	0.03	-0.04
2	Twisting SMC	4	0.01	-0.02

Table: Disturbance applied to **both areas**

Case	Controller	Settling time	Overshoot	Undershoot
1	Integral	15	0.015	-0.03
2	Twisting SMC	3	0.005	-0.01

CONCLUSION

- The results of the simulation show the convergence of ACE, which successfully includes frequency and tie-line error terms to zero and analyzed it separately.
- Twisting SMC is much better than integral controller for the case of **convergence time, overshoots and undershoots** of frequency change, tie-line power change and ACE of two area interconnected system which is validated using simulation results.



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